Statistical Analysis Plan
Study Code   CVS-2201-GLB
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A Multi-Centre, Randomised, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of CVS-221 in Adults with Elevated Cardiovascular Risk (CARDIO-SHIELD)

Sponsor: NovaCardia Therapeutics

Indication: Cardiovascular Disease Prevention (Primary Prevention in High-Risk Adults)

Phase: 3

1. Study Objectives

1.1 Primary Objective

To evaluate whether CVS-221, when added to standard-of-care lifestyle management, reduces the incidence of major adverse cardiovascular events (MACE) compared with placebo in adults at elevated cardiovascular risk.

1.2 Secondary Objectives

- To assess the effect of CVS-221 on systolic and diastolic blood pressure.
- To assess the effect of CVS-221 on total cholesterol and LDL cholesterol.
- To assess the effect of CVS-221 on body mass index (BMI) over 52 weeks.
- To characterise the safety and tolerability profile of CVS-221.

1.3 Study Design

This is a Phase 3, multi-centre, randomised, double-blind, placebo-controlled, parallel-group study. Approximately 1,800 participants will be randomised 1:1 to receive CVS-221 or matching placebo, in addition to standard lifestyle counselling, for a treatment duration of 52 weeks with an 8-week safety follow-up.

2. Eligibility Criteria

2.1 Inclusion Criteria

- Age 18 to 75 years, inclusive, at the time of screening.
- Male or female participants (gender recorded as 1 = female, 2 = male for analysis purposes).
- Body mass index (BMI) between 20 and 40 kg/m^2, inclusive.
- Systolic blood pressure (ap_hi) between 120 and 179 mmHg, inclusive, at screening (average of 3 readings).
- Diastolic blood pressure (ap_lo) between 80 and 109 mmHg, inclusive, at screening.
- Fasting total cholesterol classified as normal, above normal, or well above normal per central laboratory reference ranges (coded 1 = normal, 2 = above normal, 3 = well above normal for analysis).
- Fasting glucose classified as normal, above normal, or well above normal per central laboratory reference ranges (same 1/2/3 coding as cholesterol).
- At least one additional cardiovascular risk factor: current smoking, physical inactivity, or a first-degree relative with premature cardiovascular disease.
- Willing and able to provide written informed consent and comply with study visits.

2.2 Exclusion Criteria

- Prior myocardial infarction, stroke, or coronary revascularisation within 6 months of screening.
- Systolic blood pressure >=180 mmHg or diastolic blood pressure >=110 mmHg at screening.
- BMI <20 kg/m^2 or >40 kg/m^2.
- Known hypersensitivity to CVS-221 or its excipients.
- Current participation in another interventional clinical study.
- Pregnancy or breastfeeding at screening.
- Estimated glomerular filtration rate (eGFR) <30 mL/min/1.73m^2.
- Any condition that, in the investigator's judgement, would compromise participant safety or the integrity of study data.

2.3 Baseline Characteristics Recorded at Screening

The following variables are captured for every participant at the screening visit and form the baseline covariate set used in the efficacy and safety analyses (Table 1).

Field	Description	Coding / Units
age	Age at screening	years
sex	Biological sex	Female / Male
height	Standing height	cm
weight	Body weight	kg
bmi	Body mass index (calculated)	kg/m^2
systolic_bp	Systolic blood pressure	mmHg
diastolic_bp	Diastolic blood pressure	mmHg
cholesterol	Fasting total cholesterol category	1=normal, 2=above normal, 3=well above normal
glucose	Fasting glucose category	1=normal, 2=above normal, 3=well above normal
smoking	Current smoking status	0=no, 1=yes
alcohol	Regular alcohol consumption	0=no, 1=yes
physical_activity	Regular physical activity	0=no, 1=yes

3. Study Treatment and Administration

3.1 Investigational Product

CVS-221 is supplied as 50 mg oral film-coated tablets. Matching placebo tablets are identical in appearance, taste, and packaging.

3.2 Dosage and Administration

Participants randomised to active treatment will receive CVS-221 50 mg once daily, taken orally in the morning with or without food, for 52 weeks. Dose reduction to 25 mg once daily is permitted for participants who do not tolerate the 50 mg dose, at the investigator's discretion, following a documented tolerability assessment.

3.3 Prohibited Medications

- Other investigational lipid-lowering or antihypertensive agents.
- Strong CYP3A4 inhibitors or inducers within 14 days prior to randomisation.
- Systemic corticosteroids for more than 14 consecutive days during the treatment period.

3.4 Permitted Concomitant Medications

- Stable-dose statins, provided the dose has not changed within 8 weeks prior to screening.
- Low-dose aspirin for cardiovascular prophylaxis.
- Standard antihypertensive therapy at a stable dose.

4. Study Endpoints

4.1 Primary Endpoint

Time to first occurrence of a major adverse cardiovascular event (MACE), a composite of cardiovascular death, non-fatal myocardial infarction, or non-fatal stroke, through Week 52.

4.2 Secondary Endpoints

- Change from baseline to Week 52 in systolic blood pressure (ap_hi).
- Change from baseline to Week 52 in diastolic blood pressure (ap_lo).
- Change from baseline to Week 52 in total cholesterol category.
- Change from baseline to Week 52 in body mass index (bmi).
- Proportion of participants achieving a predicted 10-year cardiovascular risk reduction of at least 20% at Week 52.

4.3 Safety Endpoints

Incidence and severity of treatment-emergent adverse events (TEAEs), serious adverse events (SAEs), and clinically significant changes in vital signs and laboratory parameters through the end of the safety follow-up period.

5. Safety Monitoring and Visit Schedule

5.1 Adverse Event Reporting

Adverse events will be graded according to CTCAE v5.0 and recorded from the time of informed consent through 30 days after the last dose of study treatment. Serious adverse events must be reported to the sponsor's safety department within 24 hours of investigator awareness.

5.2 Stopping Rules

- Sustained systolic blood pressure >=200 mmHg or diastolic blood pressure >=120 mmHg.
- Any Grade 4 treatment-related adverse event.
- Investigator or participant decision to discontinue for safety reasons.

5.3 Visit Schedule

Visit	Timepoint	Key Assessments
Screening	Day -28 to Day -1	Eligibility, vitals, labs, ECG
Baseline	Day 1	Randomisation, baseline vitals and labs
Visit 2	Week 4	Vitals, tolerability, AE review
Visit 3	Week 12	Vitals, labs, AE review
Visit 4	Week 26	Vitals, labs, AE review
Visit 5	Week 52	End of treatment assessments
Follow-up	Week 60	Safety follow-up call

6. Statistical Analysis

6.1 Sample Size

A total sample size of 1,800 participants (900 per arm) provides approximately 90% power to detect a 25% relative risk reduction in the primary MACE endpoint, assuming a control-arm event rate of 8% over 52 weeks, at a two-sided alpha of 0.05.

6.2 Analysis Populations

- Intention-to-Treat (ITT) Population: all randomised participants, analysed as randomised.
- Per-Protocol (PP) Population: ITT participants without major protocol deviations.
- Safety Population: all participants who received at least one dose of study treatment.

6.3 Primary Analysis

The primary endpoint will be analysed using a Cox proportional hazards model, stratified by region and baseline cardiovascular risk category, with treatment group as the primary covariate. The hazard ratio and 95% confidence interval will be presented.

6.4 Handling of Missing Data

Missing data for continuous secondary endpoints will be handled using a mixed model for repeated measures (MMRM), which is robust to data missing at random. No imputation will be performed for the primary time-to-event endpoint; participants without an event will be censored at their last known event-free date.

7. References

World Health Organization. Cardiovascular diseases (CVDs) fact sheet. 2023.

Visseren FLJ, et al. 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice. European Heart Journal. 2021;42(34):3227-3337.

International Conference on Harmonisation (ICH). E9: Statistical Principles for Clinical Trials. 1998.

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